Forecasting Intraday Realized Variance with Exogenous Features and Block Ridge Regression

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Abstract

We forecast thirty-minute-ahead S&P 500 realized variance on a near-24-hour panel of over a quarter million bars (1998–2024), asking what forty-one exogenous channels add to a HAR backbone, which estimator can extract it, and what structure the nonlinearity has. Three discipline findings precede the model findings: the representation of missing and stale data moves accuracy more than any modeling decision; the back-transform from the fitting scale to the variance scale is a modeling choice, and a calibrated, strictly causal second-moment correction beats the retrospective Duan convention on every arm; and causal hyperparameter tuning on short validation tails is repeatedly worse than measured constants. The exogenous information is real but dense and weak: invisible to selection and unpenalized joint fits, recoverable only by shrinkage, and optimally handled by a two-block ridge with penalties differing by two orders of magnitude. The recoverable nonlinearity is small beside that information effect and sits at pairwise order — products of columns the panel already contains, gated on the session clock. The two-block ridge — one exact rolling solve, two fixed penalties — improves the benchmark at strong paired significance, and within the session its one-step forecast extends to the variance remaining before the close at every bar, using only what is known then, with the same advantage over the incumbent.

1 Introduction

This paper forecasts thirty-minute-ahead realized variance for the S&P 500 complex on a near-twenty-four-hour panel running from January 1998 to April 2024 — over a quarter million scored half-hour bars, most of them outside regular trading hours. The question is not whether volatility is predictable; on this panel the target's own lag ladder carries the overwhelming share of the predictable variation; the lag ladder is the benchmark throughout. The questions are what, if anything, forty-one exogenous channels — higher-order moments, liquidity, cross-sectional aggregates, sentiment, implied volatility, and options order flow — plus four scheduled FOMC calendar channels overlaying that design, add to that backbone;

which estimator class can extract it; and what structure the extractable nonlinearity has. Every answer is measured out of sample, bar by bar, under one common evaluation protocol, and every claim is paired against the same per-bar least-squares benchmark (QLIKE 0.22521).

Four findings organize the paper. The first is about the data itself. On a mixed-frequency panel, the representation of missing and stale data is a larger determinant of forecast accuracy than any modeling decision we report: at any given bar, over a third of the exogenous design is forward-filled rather than freshly printed, and a construction that cannot distinguish the two produces single-window failures orders of magnitude larger than every model-class effect in the paper. The preparation pipeline — availability indicators computed before the fill, masked scaling statistics, neutral substitution — is therefore presented in full, with the incident record attached.

The second finding is about evaluation. Models are fit on a variance-stabilized scale and scored on the raw variance scale, and the back-transform between them is not neutral: squaring a calibrated forecast is biased low by exactly the model's own residual variance, so the correction is part of each arm's forecast. We replace the standard static scalar correction with a calibrated, strictly causal retransformation protocol — a Mincer–Zarnowitz mean map re-estimated per window and an adaptive second moment seeded by the previous window and updated bar by bar — which beats the retrospective Duan [1983] convention it replaces on every arm measured, and beats it causally. A mean-reverting GARCH$(1, 1)$ refinement [**??**] of the same recursion — the covariance-stationary case, not the integrated EWMA of **?** — improves further still and is reported as an available, not adopted, upgrade.

The third finding is about the exogenous information and the estimator jointly — they cannot be separated. The exogenous channels carry real, statistically significant information, but none of it is recoverable by an unshrunk fit at this width: selection and unpenalized joint fits fail, and only shrinkage recovers the signal. The head-to-head that follows locates the structure precisely: ridge beats lasso on every design, the lasso's zeroes fall where zeroing is free, and no small set of columns or principal directions concentrates the signal. The coefficient structure is *dense but weak*, the regime of **?** and **?** — many nonzero coefficients, none large — and the estimator built for it is ridge with a block-diagonal Tikhonov penalty: the strong low-dimensional backbone barely penalized, the wide exogenous block shrunk two orders of magnitude harder, with fixed penalties and no selection step anywhere.

The fourth finding is about nonlinearity. Gradient-boosted trees beat the best linear arm on all nine feature sets, but by a margin an order of magnitude smaller than the information effect, and decompositions of the tree fits attribute the bulk of that margin to additive-plus-pairwise structure — products of columns the panel already contains, gated on the session clock. That structure is a specification statement, not yet an estimator: the paper's linear ladder ends at the block-diag ridge — one solver, two fixed penalties, refit exactly at every bar — which scores 0.22350 at DM -3.1 against the benchmark's 0.22521. The paper closes by reading the one-bar forecast against the last thirty minutes of the expiration-day chain — a last-bar same-day option trade whose position is fixed at 15:30 ET from the forecast of remaining variance against quoted implied — and shows that the sign of that gap is the tradable content: the two disagree in either direction on comparable numbers of days, a ± 1 portfolio on $\text{sign}(s)$ improves on the unconditional short, and the signal's magnitude tracks the size of the settlement move through the level of implied variance, not its direction

(Section 5.4).

A methodological theme runs through all four: discipline beats tuning. Every hyperparameter in the campaign is either a fixed constant or selected causally on a short forward validation tail, and the campaign measures what that selection buys — repeatedly, less than it costs. Tuned elastic nets lose to tuned ridge not because the data prefer dense coefficients but because penalty mistakes are cheap for ridge and destructive for selectors; a fixed forgetting half-life helps and a tuned one eliminates the gain; fixed block penalties outperform their causally tuned counterparts. The final model's constants are justified individually by measurement and never optimized on evaluation loss.

Roadmap. Section 2 places the paper in the literature. Section 3 defines the panel, the preparation pipeline, and the exogenous buckets. Section 4 fixes the evaluation framework — the benchmark, the testing machinery, and the back-transformation protocol — and then the methods: the estimators and the block constructions. Section 5 reports the results in the order summarized above. Section 6 concludes. The appendix carries the full arm-level master table and the protocol notes.

2 Related Works

Realized-volatility modeling. The paper's target and backbone descend from the realized-measure literature: realized variance as a sum of squared intraday returns [**??**], its approximate long memory documented by **?**, and the heterogeneous autoregressive model of Corsi [2009], whose daily/weekly/monthly averages we adapt to a base-2 intraday ladder. Refinements of the HAR backbone are numerous: measurement-error-dependent coefficients [**?**], state-dependent time variation [Bekierman and Manner, 2018], decoupled short- and long-run behavior [**?**], joint models of returns and realized measures [**?**], and multiple-indicator intraday structures [**?**]. We keep the backbone deliberately plain — per-bar least squares on the lag ladder — because the paper's questions concern what can be added to it, and its strength makes every addition a stringent test: on this panel the HAR backbone's coefficients are 94–97% directionally stable over two decades, so a predictor that survives next to it carries information the ladder does not.

Machine learning for volatility and return prediction. Tree ensembles are strong off-the-shelf estimators in the recent machine-learning literature on asset pricing and volatility [**??**], typically in gradient-boosted form [Ke et al., 2017, Chen and Guestrin, 2016]. Interpretability machinery — functional ANOVA, additive and pairwise additive models [Lou et al., 2013], and Shapley-based attribution [**??**] — is usually applied *post hoc*. This paper uses the trees as a specification search. The structure they recover — sparse pairwise products gated on the session clock — is written back into the linear design as explicit columns, and the distilled model is estimated in the same rolling least-squares path as every other arm.

Sparse selection versus dense shrinkage. The linear estimators are standard families — ridge [**?**], lasso [**?**], elastic net [**?**], with the least-angle-regression path [**?**] behind our exact online lasso solve — but the question they are put to has a now-substantial literature:

whether economic signal is concentrated in a few predictors or spread thinly across many. **?** compare sparse and dense representations across macro, micro, and finance prediction problems and find that the posterior rarely concentrates on a sparse model — sparsity is largely an illusion, and dense methods (ridge, factor models) predict best. **?** reach the parallel conclusion for the cross-section of expected returns, where a dense, heavily shrunk stochastic discount factor dominates sparse characteristic selections. **?** supply the estimation theory: in the weak-signal regime the zero predictor is asymptotically Bayes-optimal, ridge can improve on it while lasso cannot for any penalty, and the ranking extends to nonlinear methods (dense random forests over sparse boosted trees, ℓ_2- over ℓ_1-penalized networks); lasso's failure reflects signal *weakness*, not necessarily the absence of sparsity. The rare-and-weak detection literature [**?**] formalizes the underlying phase structure, in which a dense regime of individually undetectable effects is separated from the sparse regime by a sharp boundary. Our head-to-head (Section 5.1) is the intraday-volatility instance of this phenomenon, measured under a stricter protocol than the literature's cross-validation standard: every penalty is selected causally on a short forward tail, and the comparison is scored on a consistent loss. That protocol isolates a mechanism the asymptotic results abstract from: on short validation tails the price of a hyperparameter mistake is asymmetric across families — benign for ridge, whose loss is flat in the penalty near the optimum, destructive for selectors, whose penalty is partly a selection dial — which is why the paper's final model carries fixed penalties rather than tuned ones.

Retransformation. A mean model estimated on a log or square-root scale and scored on the original scale is not inverted by the algebraic inverse: by Jensen's inequality the naïve back-transform is biased for the original-scale mean. The standard nonparametric correction is Duan [1983]'s smearing estimator. **?** and **?** place that correction inside the forecast: under heteroscedasticity a single smearing factor is itself biased, so the retransformation is a modeling choice rather than a reporting footnote. This paper fits on a diurnally-adjusted square-root scale and scores QLIKE on raw variance, so the same gap applies. The back-transform is treated as part of the evaluation protocol — a calibrated, strictly causal second-moment correction that replaces the retrospective in-window Duan scalar — and the comparison is reported in Section 4.2.

The clock of the variance risk premium. **?** identify *when* volatility uncertainty is incorporated into real-time option prices. Volatility of volatility measured at 10:00 EST, during the U.S.–European overlap, predicts next-day variance-asset returns; that content is gone by midday. Variance prices respond with a delay, consistent with underreaction and slow-moving beliefs about volatility. This paper's last-bar option reading (Section 4.5) is the 15:30 end of the same clock. If the morning is when the early bird gets the vol, 15:30 is when that bird has already flown: quoted implied variance already exceeds the remaining-variance forecast on the bulk of expiration days, so a two-sided reading of the forecast is mostly a short, and what the forecast adds is size on that short rather than a directional flip. We do not restage their next-day VVIX regression; the object here is same-day remaining variance against the expiration-day chain.

3 Data

3.1 The Panel

The raw data are 30-minute bars for the S&P 500 complex, gridded to a regular 30-minute frequency. The panel's extent is not set by a chosen date: **a bar exists in the panel if and only if the realized-variance source printed on it** — the same observed-print fact the availability indicators of Section 3.2 are built from, applied as the grid's own existence criterion. Under this rule the panel begins at the source's first print, in January 1998, and ends at its last, in April 2024. No weekend rule, holiday calendar, or session table is supplied anywhere: weekends, holidays, early closes, and sessions that do not yet exist are all absent from the panel for the same reason — the source did not print.

The panel's session structure therefore evolves as the market's did. The source prints around the clock from its first day — the panel's first bar is an overnight bar — but the early sessions are sparser: roughly 7,500 bars per year in 1998 (median 32 bars per trading day before 2004), growing through roughly 10,600 bars in 2003 to a stable $\approx$ 12,400 bars per year from 2005 onward, when the near-24-hour cycle (Sunday 18:30 through Friday 20:00, 48 bars per full trading day, roughly 34 of them outside regular hours) is fully live. A time-of-day slot that thickens or enters mid-sample behaves exactly like a feed going live: the rolling per-slot estimators of Section 3.2 have whatever history the slot has and accumulate it causally. Any statement about intraday structure in this panel refers to the full cycle as it stood at the time, not to the 09:30–16:00 window alone.

The existence rule replaced an earlier calendar-only filter (weekend trim plus forward fill) after that filter produced a measurable incident: bars following an early close existed on the calendar grid as forward-filled dead-session bars, and on one such bar — the only one in more than twenty-five thousand — two feeds' availability indicators disagreed for a single bar, which turned a nearly singular solve into forecasts orders of magnitude off target. On the overlapping 2005–2024 span the rule removes 4,925 such dead-session bars ($\approx$ 2% of the calendar grid). The accounting is: 303,442 print-bars on the grid, 300,317 panel rows after the construction burn-in, and 273,554 scored evaluation rows after the 24,000-bar training window and the calibration burn-in of Section B.1.

The forecasting target is realized variance. For each 30-minute bar it is computed as the sum of squared 5-minute log-returns within the bar:

$$RV_{d,j} = \sum_{i=1}^{M} r_{d,j,i}^2, \qquad M = 6, \tag{1}$$

where $r_{d,j,i}$ is the i-th 5-minute log-return inside the j-th 30-minute bar of day d. Bars are indexed sequentially as RV_t. The forecast horizon is one bar ahead: the model at bar t predicts RV_{t+1}. Throughout, "realized variance" denotes the target RV_t, and "volatility" denotes its square root; models are estimated on the square-root scale (Section 3.2). The construction in Equation (1) is the standard realized-measure definition. The heterogeneous autoregressive model of Corsi [2009] uses the trading day as its base unit; here both the target and the lag ladder are defined at 30-minute resolution.

The evaluation panel. Feature construction consumes a fixed burn-in of 3,125 bars at the start of the sample (January to June 1998), a constant of the panel construction that exceeds the longest feature lookback. This reduces the grid's 303,442 print-bars to a usable panel of 300,317 bars. The walk-forward protocol consumes a further 24,000 bars as the initial training window, and the calibration burn-in of Section B.1 consumes the first evaluation window. Every arm reported in this paper is scored on the identical remaining rows; the paired bar-for-bar comparisons used throughout require this shared row index. The scored evaluation panel contains $n = 273{,}554$ bars.

Exogenous information. Beyond the target's own history, the design uses 41 exogenous channels. The channels are organised into thematic buckets that partition the columns exactly $(7 + 12 + 6 + 6 + 3 + 3 + 4 = 41)$:

- **Moments** (7): own-asset higher-order return moments: bipower variation, realized semi-variance, autocovariance, absolute, cubic, and quartic returns.
- **Liquidity** (12): trading volume, turnover (total, buy-initiated, sell-initiated), effective and quoted spreads, number of trades.
- **Market, equal-weighted** (6) and **value-weighted** (6): cross-sectional stock-factor aggregates of return moments, bipower variation, semi-variance, and absolute returns.
- **Sentiment** (3): StockTwits attention, sentiment score, and sentiment count.
- **Implied volatility** (3): VIX, VVIX, VIX3M.
- **Volatility demand** (4): options order-flow indicators (SPX open/close, all options open/close, open only).

The buckets differ in column count; liquidity contributes twelve columns and sentiment three. A per-bucket comparison therefore mixes information content with dimensionality unless it is read per feature. Availability also differs across buckets: the cross-sectional aggregates are observed on 138,357 bars, the options order-flow bucket on 54,062 bars, and the VIX family does not exist for the earliest part of the sample. Row deletion would score each bucket on a different sub-sample. Section 3.2 describes the impute-and-indicate construction used instead; it scores every bucket on the same rows.

FOMC calendar channels. Four further columns enter as a scheduled public calendar overlay on top of the 41 exogenous channels. They are not part of the $7+12+6+6+3+3+4 = 41$ partition above. The source is the `fomc release` flag: one 14:30 bar per scheduled FOMC statement (246 events; last live scheduled print 2023-11-01 14:30). `fomc_release` is 1 on that statement bar; `fomc_day` is 1 on every bar of a scheduled-statement day; `fomc_until_inv` and `fomc_since_inv` are the inverse bar-distance to the next and previous scheduled event. The unscheduled 2020-03-03 action is excluded from all four. After the last live scheduled print the four channels are missing (a dead tail), so the impute-and-indicate construction of Section 3.2 sees a stopped feed rather than a wall of zeros.

OLS on the HAR ladder plus this FOMC calendar bucket beats the baseline (HAR + calendar OLS) on the shared evaluation panel (Table 1): scheduled calendar information in the design, not a post-forecast filter. Joint OLS on all features, with the FOMC channels included among the rest, still loses to the baseline (HAR + calendar OLS).

Table 1: FOMC calendar channels on the shared evaluation panel of Section 3.1, $n = 273{,}554$. QLIKE and Diebold–Mariano t versus the baseline (HAR + calendar OLS); negative t favors the arm.

Arm	QLIKE	DM t
baseline (HAR + calendar OLS)	0.22521	+0.0
FOMC calendar bucket (OLS)	0.22400	−3.6
all features (OLS)	0.22819	+4.6
ridge (joint)	0.22551	+0.5
lasso	0.22231	−4.6
block-diag ridge	0.22206	−5.6

3.2 Preparation

This subsection specifies the preparation pipeline. The pipeline is presented in full because, measured on this panel, the representation of missing and stale data has a larger effect on QLIKE than any model-class, basis, or penalty decision reported in this paper; the magnitudes are given under "Rolling robust scaling" below. At any given bar, 36% of the exogenous design consists of forward-filled values rather than new measurements; on the sparsest channels the figure is 78%. Without availability information, an estimator cannot distinguish a filled value from a new measurement.

Overnight fills and availability indicators. The 41 exogenous channels are not all published on the near-24-hour grid. Most are US cash- or extended-session equity quantities; each prints only inside its own session. Averaged across channels, 0.638 of panel rows carry a print from the raw feed and the remaining 0.362 do not. Each channel is processed in the following order:

1. Compute a binary availability indicator from the raw feed: the indicator is 1 on bars where the feed printed and 0 otherwise.

2. Forward-fill the channel within its availability window.

3. Where the channel is structurally absent (no prior print exists), substitute a fixed in-distribution neutral value.

Every value column is accompanied by its indicator column. The indicator is computed before the fill. An earlier build of this pipeline computed the indicator after the fill, so every forward-filled bar was labelled observed. Under that build, the volatility-demand feed stopped publishing on 2023-08-31, the unlimited forward fill carried its final August print across the following eight months (8,453 bars, 3.44% of the sample), and the indicator read ≈ 1 throughout. Three further feeds (`vix`, `vix3m`, `vvix`) stopped publishing the same way on 2024-02-12. Every built cache is now gated on one check: for each channel, the mean of the constructed availability indicator must equal the fraction of rows on which the raw feed printed. The two quantities are estimates of the same number, so any gap indicates a construction error of this kind. On the uncorrected panel the check fails for 31 of 41 channels. The check requires no model, no held-out data, and no threshold tuning. We recommend it for any mixed-frequency panel in which exogenous series are filled.

The indicator also enters estimation as a regressor. Its linear coefficient absorbs any level shift between the pre- and post-availability eras, and the value column contributes only where the feed exists. Every bucket, including implied volatility and volatility demand, is therefore scored on the same 218,934-row index. The exact paired significance tests of the later sections require this shared index.

Stage 1: diurnal adjustment. Intraday volatility has a time-of-day profile. Let $s(t) \in \{1, \ldots, 48\}$ index the time-of-day slot of bar t, and let $\mathcal{W}_t = \{j : j < t,\ s(j) = s(t)\}$ be the set of the most recent $W = 20$ observations sharing that slot, with at least 5 required before the baseline is valid. For non-negative series, the baseline is the same-slot rolling mean and the adjustment divides by it:

$$\bar{RV}_{t|s} = \frac{1}{|\mathcal{W}_t|} \sum_{j \in \mathcal{W}_t} RV_j, \qquad \widetilde{RV}_t = \frac{RV_t}{\bar{RV}_{t|s}}, \tag{2}$$

Signed series (e.g. autocovariance) are divided by the same-slot rolling standard deviation instead. The window condition is $j < t$, not $j \le t$, so the baseline at bar t uses no information from bar t; the adjustment is non-anticipative. Warm-up bars pass through with baseline 1. They are consumed by the burn-in and never enter the evaluation panel.

Stage 2: the winsorized square-root target. After the diurnal adjustment, each variable class receives a fixed variance-stabilising map. The map is chosen in advance from the class's support and tail behaviour; no transform parameters are fitted. The maps are: $x \mapsto \sqrt{x}$ for squared-return and volatility-like measures, including RV itself, bipower variation, turnover, and effective spread; $x \mapsto \mathrm{sign}(x)\sqrt{|x|}$ for autocovariance; $x \mapsto x^{1/3}$ and $x \mapsto x^{1/4}$ for third and fourth moments; and $x \mapsto \log x$ for the implied-volatility family. Models are estimated on the adjusted $\sqrt{RV}$ scale. After transformation, each series is clipped to rolling 1st and 99th percentile bounds estimated on the trailing $W_{\text{clip}} = 240$ observations (about five trading days) ending at $t-1$. Winsorization is not inverted at evaluation time; only the diurnal division and the square root are inverted. Because $\mathbb{E}[\sqrt{X}]^2 \neq \mathbb{E}[X]$, squaring a forecast of $\sqrt{RV}$ gives a downward-biased forecast of RV. The standard remedy in the literature is the classical retransformation estimator of Duan [1983] — add the in-sample residual variance to the squared forecast. We refine on that traditional construction with a calibrated, strictly causal *second-order adjustment*: a Mincer–Zarnowitz mean map re-estimated at every window boundary, and an adaptive second moment that tracks the model's own recent error level bar by bar. The refinement is specified and analyzed in

Appendix B; its sensitivity against the na"ive inverse and the classical estimator is measured in Section 4.2. In raw form the back-transform is the traditional one:

$$\widehat{RV}_t^{\text{raw}} = \left(\hat{f}_t^2 + \hat{\sigma}_\varepsilon^2\right) \cdot \bar{RV}_{t|s}^{\text{baseline}}, \qquad \hat{\sigma}_\varepsilon^2 = \frac{1}{N}\sum_i (y_i - \hat{y}_i)^2. \tag{3}$$

HAR features. The target's own history enters through trailing rolling means of the adjusted series on a base-2 geometric ladder,

$$\mathcal{L} = \{1,\ 2,\ 4,\ \ldots,\ 2^{k_{\max}}\}, \qquad \overline{RV}_t^{(k)} = \frac{1}{k}\sum_{i=0}^{k-1} \widetilde{RV}_{t-1-i}, \tag{4}$$

The rungs span roughly 30 minutes, 2.5 hours, half a trading day, 2.6 days, 13 days, and 65 trading days. The sum starts at $t-1$, so the feature at bar t contains no contemporaneous information. The ladder adapts the daily/weekly/monthly averages of Corsi [2009] to six intraday horizons. The same ladder is applied to each included exogenous channel, contributing $|\mathcal{L}| = 6$ rolling-mean columns per channel.

Rolling robust scaling. Input features to the linear estimators are standardised by subtracting a rolling median and dividing by a rolling interquartile range. Both statistics are estimated from the current training buffer and updated at every step. Two rules govern the estimation.

First, the median and IQR are estimated from observed rows only, i.e. rows on which the availability indicator is 1. If the statistics are instead estimated over forward-filled values, a series whose feed has stopped contributes identical repeated values, its rolling IQR shrinks toward zero, and the standardising division divides by a value near zero. On the uncorrected panel, the stale volatility-demand columns reached 2,260 rolling IQRs by this mechanism; the raw series did not move, and the denominator shrank.

Second, a scale is used only after 512 observed values have accumulated, and the IQR is floored at 1% of the largest scale the channel has shown. This rule exists because masked estimation fails at the start of a feed's life: with few observed rows, a low-dispersion opening window would otherwise set the scale for all subsequent bars.

The three elements — indicators computed before the fill, masked estimation, and neutral substitution — are applied together. Each element applied alone does not remove the failure. Applied together, they reduce the worst channel's maximum standardised value from 2,260.4 to 96.2 rolling IQRs. On the October 2023 window affected by the stale feed, the uncorrected panel scored QLIKE 6.37170 and the repaired panel scored QLIKE 0.12391. Every modelling effect reported in this paper lies between 0.0005 and 0.004 QLIKE. This difference in magnitude is the reason the preparation pipeline is presented before the modelling sections.

All stages are strictly causal: every quantity dated t uses only information available strictly before t.

3.3 Descriptive Analysis

4 Methods

The methods come in two halves. The first half fixes the evaluation framework before any comparison is made: the comparator each arm is measured against and the testing machinery that accompanies it (Section 4.1); and the inheritance rule that makes the whole stack binding on every downstream comparison (Section 4.2). The back-transformation from the fitting scale to the variance scale, and the calibrated, strictly causal second-order adjustment it carries, are defined in Appendix B and summarized in Section 3.2. The second half defines what is compared: the estimators — one unpenalized member, the penalized linear family, and two tree ensembles — with the walk-forward protocol and the causal tuning rule every tuned arm obeys (Section 4.3); and ridge with a block-diagonal Tikhonov penalty (Section 4.4). Every estimator is fit on the shared panel of Section 3 and subject to the hyperparameter discipline of Section 4.1: no quantity is selected on evaluation loss.

4.1 The OLS–HAR Benchmark

Every result in the remainder of this paper is referenced to a single fixed comparator: ordinary least squares on the six HAR features of Equation (4) plus calendar dummies, with no exogenous variables, refit at every bar. The forecasting protocol is:

1. Walk-forward with a rolling origin. The training window is the most recent $T_{\text{train}} =$ 24,000 bars (approximately 500 trading days at 48 bars per day). The window slides forward by one bar per prediction.

2. The model is refit at every bar. The refit is exact and computationally feasible because the sliding-window normal equations admit rank-1 up-dating (add the entering bar) and down-dating (remove the exiting bar).

3. The benchmark is read from the ridge path at $\alpha = 0$; it is therefore exact OLS.

4. Each forecast is mapped back to raw variance through Equation (3).

5. Forecasts are scored by QLIKE [Patton, 2011],

$$L_{\text{QLIKE}} = \frac{1}{N} \sum_{t=1}^{N} \left[r_t - \log r_t - 1 \right], \qquad r_t = \frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}}, \tag{5}$$

computed on the full 273,554-bar evaluation panel. QLIKE is used for three reasons: it is scale-invariant in the level of volatility; it is strictly consistent for the conditional variance; and it penalises under-prediction more heavily than over-prediction, which matches the direction in which forecast errors are more costly for this target. QLIKE values are not comparable across preprocessing regimes: on the diurnally adjusted panel of the reference study, the same class of models scores several tenths of a QLIKE unit lower than on a raw, unadjusted panel. Every comparison in this paper is therefore made within one preprocessing regime, and every table is labelled with the panel it was computed on.

Table 2: The benchmark: per-bar OLS on the HAR ladder plus calendar dummies, scored on the frozen panel under the evaluation protocol of Sections B.1–4.2. QLIKE is computed on the raw variance scale; the secondary metrics are computed on the fitting (square-root) scale where the model is estimated, on the same scored pairs.

Scored bars	273,554 (positivity exclusion removes 0.2%)
QLIKE (raw scale)	0.22521
Out-of-sample R^2 (fit scale)	0.417
Mincer–Zarnowitz α (fit scale)	−0.034
Mincer–Zarnowitz β (fit scale)	1.039
MSE (fit scale)	0.1397
MAE (fit scale)	0.2208

Squared-error metrics computed on the raw variance scale are uninformative on this panel: the median absolute error is 3.3×10^{-7} against a maximum of 35.6, so a handful of volatility-spike bars dominates every squared-error aggregate — the asymmetry that motivates QLIKE as the primary loss (Section 4.1).

Differences in QLIKE at the fourth decimal place require a significance assessment. Every comparison against this benchmark is therefore accompanied by a Diebold–Mariano test [Diebold and Mariano, 1995] on the per-bar loss differential. The test procedure is:

1. Compute the per-bar QLIKE loss for each of the two arms on the shared row index, and take the per-bar difference.

2. Estimate the long-run variance of the loss differential with a Newey–West estimator.

3. Apply the small-sample correction of Harvey et al. [1997] to the test statistic.

Where several arms are compared simultaneously, we report the Model Confidence Set of Hansen et al. [2011]. Every arm is scored on the identical row index, so all of these tests are paired bar-for-bar comparisons.

Hyperparameter discipline. No quantity anywhere in this paper is selected using evaluation loss. Every hyperparameter that is tuned is tuned *causally*: the penalty (or penalty vector) is re-selected periodically from a grid fixed in advance, by minimizing validation error on a forward split of the current training window — fit block, embargo, validation tail — so that no forecast's configuration uses information from that forecast's future. Every hyperparameter that is not tuned is a fixed constant, set before the campaign was scored and reported with its arm: the fixed-penalty arms exist precisely as controls for what tuning contributes, and the structural constants of the design (training-window length, re-selection cadence, winsorization quantiles, block widths, estimation-window lengths) are conventions of the panel and estimator construction, disclosed where they are defined and never optimized against any loss.

4.2 A Common Evaluation Protocol

Every forecast in this paper passes through Equation (17) before any QLIKE number is computed. Every downstream comparison — the linear-class comparisons of Section 5.1, and the tree and final-model results of Sections 5.2 and 5.3 — scores its arms on raw-scale pairs produced by exactly this map, with exactly these window and residual choices.

Because the calibration pair and the second-moment coefficients are model-specific, the correction does not shift all arms by a common constant. It inflates the forecasts of an arm with larger residual variance by more, and it straightens the forecasts of a miscalibrated arm by more. Under QLIKE's asymmetric penalty, this interacts with each arm's calibration in a way that a shared correction would not. A comparison run under a different convention — an uncalibrated squared forecast, a a scalar classical estimator, or no correction at all — would be a comparison of different forecast objects, and it would not necessarily induce the same ranking.

That comparison is run, restricted to exactly three conventions on the identical persisted per-bar arrays and the identical exclusion rule. Convention *none* is the naïve inverse of Section B.1: the squared forecast with the diurnal adjustment reversed, $\widehat{RV}_t^{\mathrm{raw}} = \hat{y}_t^2 B_t$, with no second-moment term. Convention *Duan* is the classical Duan [1983] estimator — the in-window scalar special case identified after Equation (17) — with one scalar per evaluation window, the mean of that window's own squared fit-scale residuals, added to the squared forecast before the baseline reversal. Convention *second-order adjustment* is the calibrated back-transformation of Equations (13)–(17), the convention every headline number in this paper inherits. Table 3 reports each completed arm's pooled QLIKE under the three conventions, with within-convention ranks. The rank-stability summary is the Kendall rank correlation between the second-order-adjustment ranking and each alternative's: $\tau = 0.682$ against *none* and $\tau = 0.859$ against *Duan*.

Table 3: The benchmark under the three back-transform conventions. The full per-arm table is Appendix Table 9; Kendall τ of each alternative ranking against the second-order-adjustment ranking over all arms: none 0.682, traditional Duan 0.859.

	No correction	Classical Duan	Second-order adjustment
OLS on HAR + calendar (benchmark)	0.24097	0.23165	0.22521

The correction is treated as part of the evaluation protocol: it is frozen here, before the first comparison, and no downstream section revisits it. Any future change to the second-moment model — including the learned correction contemplated for this slot — would constitute a change of protocol, and would require re-scoring every comparison that inherited the old one.

4.3 Estimators and Causal Tuning

Minimum-norm least squares. The unpenalized estimator is *minimum-norm least squares*: among all coefficient vectors that minimize the training-window sum of squares, the unique one of minimal ℓ_2 norm, computed through the pseudoinverse of the window gram at the standard machine-precision cutoff. This is the $\alpha \to 0^+$ limit of ridge regression, it is defined at any design rank, and it carries no penalty and no tuned quantity of any kind — the strongest attribution claim an unpenalized estimator supports. The rank cutoff is not a tuned quantity either: the centered gram's spectrum is gapped by more than five orders of magnitude around the retention threshold (on the widest design, largest discarded eigenvalue 1.8×10^{-11} relative, weakest retained 6.2×10^{-6}, with no eigenvalue between them at any bar), so every tolerance across four decades retains the identical subspace and reproduces the reported forecasts bitwise. Each arm is refit at every bar over a rolling window of 24,000 bars (500 trading days). On a full-rank design (the benchmark's 52 columns) minimum-norm least squares coincides exactly with ordinary least squares. An earlier pass of this campaign used the plain normal-equations solve, whose solution is arbitrary in near-null directions of the design; two single-window incidents in the campaign record trace to that path, and its results are archived. Because the pseudoinverse is defined at any rank, the estimator requires no rank guards; the only design intervention retained is the go-live participation rule of Section 3.2, which is a statement about data existence, not numerics.

Estimators. The design matrix X has one row per bar in the current training window and one column per feature: the HAR ladder of the target, the calendar features, and the full exogenous block of 523 columns. Let y be the vector of targets over the same window, and let x_s denote row s of X. The two estimators solve, at each refit,

$$\hat{\beta}^{\text{ridge}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^\top \beta\right)^2 + \alpha \, \|\beta\|_2^2,$$
$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^\top \beta\right)^2 + \lambda \, \|\beta\|_1,$$

where the sums run over the bars of the training window and the penalties α and λ are held constant between the periodic re-selections described below. The ℓ_1 penalty produces a coefficient vector supported on a subset of columns; the ℓ_2 penalty produces a coefficient vector with all entries nonzero. The elastic net [?] interpolates the two,

$$\hat{\beta}^{\text{enet}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^\top \beta\right)^2 + \lambda \left(\tfrac{1-\rho}{2} \, \|\beta\|_2^2 + \rho \, \|\beta\|_1\right),$$

recovering ridge at $\rho = 0$ and the lasso at $\rho = 1$. Its role in this paper is diagnostic as much as predictive: in the causally tuned arms the mixing share ρ is *free* — re-selected from $\{0.25, 0.5, 0.75, 1.0\}$ jointly with the penalty strength at every retune — so the estimator, not the author, chooses how much selection to apply, and the selected trajectory is part of the evidence in Section 5.1. All three penalized families are parameterized in the units of `scikit-learn`'s solvers, against which every custom solver below was validated.

Walk-forward protocol. The panel is computed as 100 walk-forward chunks over a rolling 500-trading-day (24,000-bar) training window. Every arm is refit *every bar*: at each bar the oldest bar leaves the window, the newest enters, and the coefficients are re-estimated before the next forecast is issued. Each arm re-selects its own penalty periodically and causally: every 250 solves, the current training window is split forward into a fit block, a

25-bar embargo (covering the longest moving average entering the features), and a 125-bar validation tail, and the penalty value minimizing validation MSE on that tail is adopted. No forecast's hyperparameters ever see its own future. The lasso and elastic-net arms are solved as exact every-bar online homotopies [?]: the solution and active set at each bar are obtained by exactly updating the previous bar's solution for the one-row change in the window, not by a warm-started batch solver terminated at a tolerance, so the head-to-head is not confounded with optimization error. The implementation is a re-authored recursive elastic net (Gram-form active-set homotopy with rank-1 Cholesky maintenance, validated coefficientwise against `scikit-learn`). The ridge arm runs on an exact rank-one update path.

Tree ensembles. The nonlinear members of the estimator set are two implementations of gradient-boosted regression trees. Gradient boosting fits an additive ensemble forward-stagewise: starting from a constant, at stage m it adds a shallow regression tree f_m fit to the negative gradient of the loss at the current ensemble's predictions,

$$F_m(x) = F_{m-1}(x) + \eta\, f_m(x), \qquad f_m \approx \arg\min_{f\in\mathcal{T}} \sum_i \Big(-\frac{\partial \ell(y_i, F_{m-1}(x_i))}{\partial F_{m-1}(x_i)} - f(x_i) \Big)^2,$$

where $\mathcal{T}$ is the space of depth-limited regression trees, $\eta \in (0, 1]$ is the shrinkage rate, and for the squared loss used throughout this paper the negative gradient is exactly the current residual, so each tree is a regression tree on the residuals of the ensemble so far.

XGBoost [Chen and Guestrin, 2016] fits each tree to a second-order Taylor expansion of the loss rather than the raw gradient. Writing $g_i = \partial\ell/\partial F_{m-1}(x_i)$ and $h_i = \partial^2\ell/\partial F_{m-1}(x_i)^2$, the per-tree objective

$$\mathrm{Obj}_m \;\approx\; \sum_i \Big[g_i f_m(x_i) + \tfrac{1}{2} h_i f_m(x_i)^2 \Big] + \gamma T + \tfrac{1}{2}\lambda \sum_{j=1}^{T} w_j^2,$$

with T leaves and leaf weights w_j, has closed-form optimal weights $w_j^* = -G_j/(H_j + \lambda)$, where G_j and H_j are the gradient and Hessian sums over the rows in leaf j, and a split is accepted only if it improves $\frac{1}{2}\big[G_L^2/(H_L + \lambda) + G_R^2/(H_R + \lambda) - G^2/(H + \lambda)\big]$ by more than γ. The curvature term makes the leaf weights loss-aware (for squared error $h_i = 2$ and the scheme reduces to a regularized mean), and γ prices complexity per added leaf explicitly; row and column subsampling decorrelate successive trees. Our configurations use its histogram split finder (`tree_method="hist"`).

LightGBM [Ke et al., 2017] uses the first-order scheme with three engineering choices that matter at this design's width. First, feature values are binned into histograms, so split search costs depend on the bin count rather than the 24,000-row window length. Second, gradient-based one-side sampling keeps all large-$|g_i|$ rows and samples small-$|g_i|$ rows with compensating weights, concentrating the fit on the hard bars. Third, exclusive feature bundling merges mutually exclusive columns — a property this design has in abundance, since an availability indicator and its siblings are zero whenever another feed's indicator is one — shrinking the effective width without changing the fitted function class. Trees grow leaf-wise (best-first) under a leaf-count cap rather than level-wise: deeper trees for the same budget, with the cap standing in for depth control.

Roles in the campaign. The two families enter twice, under different tuning disciplines. In the causal-tuning battery they are tuned arms like any other: their hyperparameters are re-selected on the same periodic forward-split schedule as every tuned arm, and the battery fully crosses the five estimators of this subsection with the nine feature buckets on the battery's shared panel (n = 218,934 thirty-minute bars, identical rows in every cell). In the frozen *expert bank* the configurations are fixed permanently before any scored bar: a dev-span study (rows before 2003, screen refits every 250 bars) searched each family's box — LightGBM: `num_leaves` 15–255 and `min_child_samples` 20–500 and `lambda_l1/l2` 10^{-8}–10 on log scales, `learning_rate` 0.01–0.3, `n_estimators` 50–500, feature and bagging fractions 0.5–1.0; XGBoost: `max_depth` 3–10, `min_child_weight` 1–100, `reg_alpha/reg_lambda` 10^{-8}–10, `gamma` 10^{-8}–1 on log scales, the same learning-rate and estimator ranges, subsample and column-subsample 0.5–1.0 — and the menu froze the top eight unique configurations plus two seeded Sobol points per family (2×10 experts). Each expert is then refit on the trailing 24,000-bar window at the disclosed refit cadence, on the identical wide design the linear arms read, with a fixed seed and threads following the slot count, so the tree class and the linear class are compared on the same information set with no per-arm tuning freedom. The shared comparator in both roles is the per-bar OLS–HAR incumbent (Section 4.1).

4.4 Block-Diagonal Tikhonov

A single ridge penalty cannot serve the HAR backbone and the wide exogenous design at once. Penalizing the HAR-plus-calendar design alone raises QLIKE relative to plain OLS (ridge $t = +5.0$ against the incumbent): those columns are few, well-conditioned, and individually strong, so any shrinkage on them is pure bias. The wide exogenous design requires heavy shrinkage — that is the head-to-head's result. A scalar α large enough for the weak exogenous columns over-shrinks the HAR columns; a value small enough for the HAR columns under-shrinks the exogenous columns.

The resolution is a **block-diagonal Tikhonov** penalty [**???**]: one jointly solved least-squares problem, blocks ordered from strong and low-dimensional to weak and wide, each shrunk harder than the last. The B-block problem

$$\hat{\beta} = \arg\min_{\beta_1,\ldots,\beta_B} \left\| y - \sum_{b=1}^{B} X_b \beta_b \right\|_2^2 + \sum_{b=1}^{B} \alpha_b \|\beta_b\|_2^2, \qquad \alpha_1 < \alpha_2 < \cdots < \alpha_B,$$

reduces to ordinary unit-penalty ridge by column scaling. With $\tilde{X} = \left[\, X_1/\sqrt{\alpha_1} \;\cdots\; X_B/\sqrt{\alpha_B} \,\right]$ and $\gamma_b = \sqrt{\alpha_b}\,\beta_b$,

$$\left\| y - \sum_b X_b \beta_b \right\|_2^2 + \sum_b \alpha_b \|\beta_b\|_2^2 = \|y - \tilde{X}\gamma\|_2^2 + \|\gamma\|_2^2,$$

so $\hat{\beta}_b = \hat{\gamma}_b / \sqrt{\alpha_b}$. After scaling, every nested design runs in the same exact rank-one rolling least-squares path as every other linear model in this paper, refit at every bar. Setting a trailing block's coefficients to zero recovers the previous design exactly. There is no selection step and no tuned dial: the penalties are fixed constants.

Two blocks. The two-block design uses $B = 2$ and $(\alpha_1, \alpha_2) = (1, 100)$. Block 1 ($X_1$) is the HAR ladder of the target. Block 2 (X_2) is the same HAR averaging construction applied to every exogenous feature — HAR-of-`all_features`, the full wide basis. The two blocks are not fit separately and stacked. Setting $\beta_2 = 0$ reduces the model to ridge at $\alpha = 1$ on the backbone, which differs from the OLS–HAR incumbent of Section 3 only by that nominal penalty. The estimator matches the measured coefficient structure: block 2 keeps every column and shrinks them two orders of magnitude harder than block 1.

4.5 A Last-Bar Same-Day Option Reading

The one-bar forecast can be read against a listed price on the last thirty minutes of an expiration day. What follows fixes the instrument, the two measurements, the position rules, and the control; constants were set before the portfolio was scored.

Instrument. S&P 500 weekly (SPXW) options with zero days to expiry, cash-settled on the day's official close. The panel is the expiration-day chain at 15:30 ET (bid, ask, mid, underlying, quoted implied volatility), 2020 through 2025. At 15:30 the trade is the nearest out-of-the-money pair: the smallest listed strike $K_c \geq S$ with a live mid and the largest listed strike $K_p \leq S$ with a live mid, where S is the 15:30 underlying print. When S sits on a listed strike the two coincide (a straddle); otherwise the package is a one-strike-wide strangle. Entry $P_{15:30}$ is the sum of the two mids. The position is held to the close and settled at intrinsic against the official S&P 500 close S_{cl},

$$\Pi_{\text{long}} \; = \; \big(S_{\text{cl}} - K_c\big)^+ + \big(K_p - S_{\text{cl}}\big)^+ \; - \; P_{15:30}, \tag{6}$$

with no delta hedge and no share unwinding. The long return is $R = \Pi_{\text{long}}/P_{15:30}$; a short earns $-R$. By construction $R \geq -1$, with equality on the days the close settles strictly between K_p and K_c and both legs expire worthless. The 16:00 chain underlying is retained only as a check on the settlement print. Half sessions without a close are dropped.

Forecast remaining variance. Seven of the paper's per-bar forecasts enter the same rules on the same contracts: the baseline (HAR + calendar OLS) forecast (Section 4.1), the block-diag ridge (HAR versus exogenous block, separate penalties; Section 4.4), LightGBM and XGBoost on the wide all-features design (Section 5.2, frozen expert menu), a causally tuned lasso on that same design under the selection protocol of the paper's tuned ridge (Section 5.1), a lasso at a fixed $\alpha = 10^{-4}$, and a causally tuned elastic net — the interpolating ℓ_1/ℓ_2 family — on the same wide all-features design (battery grid `logspace(-6,-2,5)` at `l1_ratio=0.5`, arm `b3_enet_tuned`). All seven are stored on the estimation scale $y = \sqrt{RV/B}$. The 15:30 row is mapped to raw thirty-minute variance by the causal second-order back-transform of Section 4.2: a Mincer–Zarnowitz map $m = a + b\,\widehat{y}$ and a residual second moment $\hat{\sigma}^2$, both fit on the previous 250 days only, then $\widehat{RV} = (m^2 + \hat{\sigma}^2)B$, an estimate of $E[RV]$ for the 15:30–16:00 bar. Forecast files end on 30 April 2024, so later expiration days are not scored.

Two conventions matter for honesty. *Alignment*: forecast rows are labelled by the end of the bar they predict, so the row read at 15:30 is the one stamped 16:00 — issued at 15:30 for the thirty minutes that remain — and not the stamp-15:30 row, which was issued at 15:00 for a bar already finished; every input to the signal below is therefore known at 15:30. *Weighting*: the map and the residual moment solve weighted least squares with weights $1/\max(\widehat{y}_u,\, q_{10})^2$, where q_{10} is the tenth percentile of $\widehat{y}$ within the same trailing window, and the fit window is restricted to regular-session bars. The weighting addresses a level effect — an unweighted fit is dominated by the largest bars — and was adopted on the forecast-accuracy referee (QLIKE improves for all seven models), not on any portfolio outcome.

Quoted implied volatility. Implied volatility is the chain's quoted value on each of the two legs at 15:30; the package quote is the mean of those two. The vendor quote is an *hourly* volatility while the window that remains is thirty minutes, so the quote is rescaled before it is compared with anything,

$$\mathrm{IV}_{30,t} \;=\; \mathrm{IV}_t^{\text{hourly}}\big/\sqrt{2}, \tag{7}$$

and $\mathrm{IV}^2_{30,t}$ is the implied variance of the remaining bar — the same window, and the same units, as $\widehat{RV}_t$. No separate inversion from the mid is applied; as a check, inverting the Black–Scholes–Merton price of the package mid over the remaining half hour reproduces the vendor quote to quote precision, so "quoted" and "inverted" implied volatility coincide on this panel.

Signal. The two measurements are compared in variance space,

$$s_t \;=\; \widehat{RV}_t - \mathrm{IV}^2_{30,t}, \qquad \mathrm{VRP}_t \;=\; \mathrm{IV}^2_{30,t} - \widehat{RV}_t \;=\; -\,s_t, \tag{8}$$

so $s_t > 0$ says the forecast is above the price of the remaining variance and $s_t < 0$ says the premium is positive. The daily portfolio return is $R'_t = q_t R_t$ for a position q_t fixed at 15:30 from information dated t or earlier.

Position rules. Two rules are scored on the same R_t and the same days; only q_t changes.

- *Short volatility*: $q_t = -1$ every day. The rule takes no forecast — it is the variance-risk-premium portfolio with nothing in it, and the control for the other.
- $\mathrm{sign}(s)$: $q_t = \mathrm{sign}(s_t)$, long the package when the forecast exceeds implied variance and short otherwise. Direction only, at unit size. This is the headline rule.

Compounding at a fraction of wealth. The tables above price one unit of premium per day and add P&L up. A reinvesting trader bets a fraction f_t of current wealth instead, so wealth compounds as $W_T = \prod_t (1 + f_t\, q_t R_t)$ and the natural objective is the annualized mean of $\log(1 + f_t R'_t)$. Ruin is one bad day: any $f \geq 1/|\min_t R'_t|$ puts wealth at or below zero on the worst day, so the unhedged short portfolio, with a worst day near -10 premium units, can only ever bet a small fraction of wealth no matter how good its mean. A defined-risk variant, which would bound the worst day by construction, is parked and not reported here. The fraction is estimated causally from the portfolio's own past returns,

$$\hat{f}_t \;=\; \min\left(\frac{\hat{\mu}_{t-1}}{\widehat{E[R'^2]}_{t-1}},\; \frac{1}{|\min_{u \leq t-1} R'_u|} \right)_+, \tag{9}$$

the quadratic growth rule under a running ruin cap, expanding with a minimum of 63 sessions and lagged one day. The half-fraction path $\hat{f}_t/2$ is scored alongside, and every realized wealth factor is asserted positive.

"A fraction of wealth" needs a unit before any growth number can be read, and two frames are used. In the *per-premium* frame — the frame for the plain portfolio — f_t is the share of wealth deployed as body premium; a bought straddle's premium is also its capital at risk, while the uncapped short side's capital at risk is unbounded, which is exactly why the ruin bound above governs the admissible fraction. A true *capital-at-risk* frame — f_t posted as max-loss collateral, the day's return on that capital bounded below by -1 by construction — exists only once the worst day is bounded; it applies to a defined-risk variant that is parked and not reported here. One unit of wealth posted as collateral controls several times less premium exposure than one unit deployed as premium, and the ratio varies day to day, so growth rates compare only within a frame, never across.

Scoring and control. The unit of account is the expiration day. For a portfolio of n daily returns, $t = \sqrt{n}\,\overline{R'}/\hat{s}(R')$, and the annualized Sharpe ratio is $\overline{R'}/\hat{s}(R') \times \sqrt{252}$; the two carry the same information at fixed n, and both are reported so the tables read either way. Fills are at the 15:30 mid throughout, and no spread crossing, borrow, or market-impact model is applied.

The short-volatility control is the 15:30 end of the clock in Section 2: **?** locate the predictive content of volatility uncertainty at 10:00 EST and gone by midday; by the last bar the quoted implied already embeds a premium over remaining realized variance. On the scaled comparison of Equation (7), implied variance exceeds the forecast on about 61% of days, so the portfolio leans short more often than long. What the forecast contributes at this hour is the *sign* of the day — which side of the quoted premium to stand on. Sizing beyond the sign is tested rather than assumed: Section 5.4 reports the magnitude experiments, none of which is retained; the only graduation of exposure kept is the causal fraction-of-wealth rule of Equation (9), which reads the portfolio's own past returns, not the signal.

4.6 Implied Volatility by Root Finding

Where an option price is read as a volatility, the map is the Black–Scholes–Merton formula [**??**]. For a European option on a forward-like underlying S with strike K and time to expiry τ, at zero rate,

$$V^{\mathrm{BSM}}(\sigma) \;=\; \phi\big[\, S\,\Phi(\phi d_1) - K\,\Phi(\phi d_2)\,\big], \qquad d_{1,2} \;=\; \frac{\log(S/K) \pm \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \tag{10}$$

with $\phi = +1$ for a call and -1 for a put and Φ the standard normal distribution function. The implied volatility of a quoted price V^* is the root of $V^{\mathrm{BSM}}(\sigma) - V^* = 0$. The map $\sigma \mapsto V^{\mathrm{BSM}}(\sigma)$ is continuous and strictly increasing on $\sigma > 0$ — its derivative is the vega $S\,\varphi(d_1)\sqrt{\tau} > 0$ — from the intrinsic value $(\phi(S-K))^+$ at $\sigma \to 0$ to S (call) or K (put) as $\sigma \to \infty$, so a root exists and is unique exactly when V^* lies strictly inside that interval, and there is no closed form for it. It is found numerically: a bracketed bisection on $[10^{-4}, 5]$ that halves the interval thirty times, followed by three Newton steps $\sigma \leftarrow \sigma - (V^{\mathrm{BSM}}(\sigma) - V^*)/\mathrm{vega}(\sigma)$

from the bisection point. Bisection supplies a guaranteed bracket; Newton supplies the last digits. Prices at or below intrinsic — deep in-the-money quotes on a same-day expiry, and any quote at $\tau = 0$ — have no root and are excluded rather than clipped. The solver is vectorized over contracts and agrees with a scalar Brent root-finder to 5×10^{-7} on a random battery.

5 Results

The results proceed from estimator comparisons to nonlinear structure. Section 5.1 establishes the coefficient structure of the exogenous signal — dense but weak. Section 5.2 locates the recoverable nonlinearity. Section 5.3 reports the block-diagonal Tikhonov.

5.1 Lasso versus Ridge

The exogenous panel carries information beyond the HAR lag structure of Corsi [2009]; whether any of it is recovered is decided by the estimator. This subsection holds the information set fixed and varies only the penalty of the linear estimator. Two estimators are compared: the lasso [**?**], which applies an ℓ_1 penalty and sets a subset of coefficients to exactly zero, and ridge [**?**], which applies an ℓ_2 penalty and shrinks all coefficients toward zero without setting any to zero. The result of the comparison is a property of how the predictive coefficients are distributed across the columns of the design. In this section that property is measured directly: ridge attains lower out-of-sample QLIKE than the lasso on the full design (Section 5.1); the lasso zeroes 304 of 523 exogenous columns on the full basis, of which 235 are dead or constant availability indicators (Section 5.1); and five principal components retain 72% of the feature variance while losing essentially all of the predictive signal (Section 5.1). The coefficient structure is *dense but weak*, in the sense of the sparse-versus-dense prediction literature [**???**]: many nonzero coefficients, none individually large. Section 4.4 defines the estimator built for this structure: ridge with a block-diagonal Tikhonov penalty whose first design uses $\alpha = 1$ on the HAR-of-target block and $\alpha = 100$ on the HAR-of-all-features block, with no selection step.

The head-to-head.

Design. The two penalized linear estimators are run on the full feature design — every exogenous column at once — on one shared out-of-sample panel of $n = 218{,}934$ thirty-minute bars, with identical rows in both cells; nothing varies between the two arms except the penalty.

Comparison protocol. Both arms are compared against the shared incumbent of Section 3: per-bar OLS on HAR plus calendar features with no exogenous inputs, at the benchmark QLIKE of 0.22521. For each arm, let ℓ_t^{arm} and ℓ_t^{inc} be the per-bar QLIKE losses [Patton, 2011] of the arm and the incumbent at bar t, and let $d_t = \ell_t^{\text{arm}} - \ell_t^{\text{inc}}$. The Diebold–Mariano statistic [Diebold and Mariano, 1995] is the sample mean $\bar{d}$ divided by its Newey–West HAC standard error, with the small-sample correction of Harvey et al. [1997] applied; a negative t favors the arm.

Result. The comparison is run on the full design — the HAR ladder, the calendar features, and the full exogenous block — with both estimators under the same causal tuning protocol, so that neither family carries an informed penalty choice the other lacks (Table 4). Under the evaluation protocol, causally tuned ridge (0.22452, DM -1.1) beats the incumbent; causally tuned lasso (0.22566, DM +0.5) does not. Keeping every column and shrinking produces a lower loss than selecting a subset, when both families choose their penalty by the same causal rule. The elastic net (causally tuned) on the same full design scores 0.22464 (DM -0.6), interpolating ridge and lasso, closer to ridge, and not distinguishable from the incumbent at conventional size.

Table 4: Ridge versus lasso versus elastic net on the full design under the evaluation protocol, out-of-sample QLIKE with the Diebold–Mariano t-statistic against the OLS–HAR incumbent in parentheses. Panel: the frozen panel of Section 3, $n = 273{,}554$ identical rows. Ridge is the fine-grid causal tune already harvested on this root (`b1_ridge_tuned_fine`); lasso is `b2_lasso_tuned`; elastic net is `b3_enet_tuned`. Negative t favors the arm.

	Ridge	Lasso	Elastic net
Causally tuned	0.22452 (-1.1)	0.22566 (+0.5)	0.22464 (-0.6)
Incumbent (per-bar OLS–HAR)		0.22521	

A penalty-level sweep locates where the tuned arms operate. Holding the ridge family fixed and varying only the constant, QLIKE falls monotonically in α across the tested grid: 0.23219 at $\alpha = 0.1$, 0.23159 at 0.3, 0.23083 at 1, 0.23003 at 3, and 0.22909 at 10, still decreasing at the grid's edge; the tuned-ridge arm, which re-selects from a grid reaching $\alpha = 1000$, lands below every fixed point. The penalty *level* moves the loss by more than the penalty *family* does at any fixed level, and the wide design rewards heavier shrinkage than the $\alpha = 1$ default — the direction the block penalties of Section 4.4 take further. One caveat: all DM statistics are computed against the incumbent, a third model, so the ridge-minus-lasso differences themselves carry no significance statement yet.

The diagnosis: dense, not sparse. The lasso's own solution path shows what ℓ_1 selection does on this design. On the full basis the lasso zeroes 304 of 523 exogenous columns. Of those 304, 235 — 77% — are dead or constant availability indicators. Most of the lasso's selection therefore removes columns that carry no signal. Removing such columns does not lower QLIKE relative to ridge, because ℓ_2 shrinkage makes the same columns small without making them zero. Where the ℓ_1 penalty zeroes live columns, QLIKE rises: the margin by which ridge beats the lasso is carried by the live columns the lasso zeroes, not by the dead ones.

A principal-component check measures whether the spread of coefficients across columns is a low-dimensional structure in a rotated basis. Truncating the design to five principal components retains 72% of the feature variance and loses essentially all of the predictive signal. The signal therefore does not lie in the high-variance directions of the feature covariance. The measured structure is: (i) no small subset of columns concentrates the signal — the only columns the lasso can zero without raising QLIKE are dead or constant ones; and (ii) no small set of principal directions concentrates it. The coefficient structure has many small nonzero entries in the coordinate basis and is not low-rank in the data's covariance geometry. This is the dense-but-weak structure of **?**, verified on the solution path rather than inferred from the loss ordering.

The block-diagonal Tikhonov of Section 4.4 is the estimator built for this structure; its scores are in Section 5.3.

5.2 Gradient-Boosted Trees

The linear results are now in place: the exogenous signal is dense and weak (Section 5.1). The remaining question is what nonlinear structure adds. First, gradient-boosted trees beat the best linear arm on every feature set, by a small margin. Second, decompositions of the tree fits attribute the bulk of that margin to additive-plus-pairwise structure, and the pairwise part to products of columns the panel already contains, gated on the session clock.

The expert bank. The twenty frozen configurations of Section 4.3 were walked per bar on the full panel and scored under the evaluation protocol. Seventeen of the nineteen completed experts beat the benchmark, at QLIKE 0.21796–0.22108 against the benchmark's 0.22521 (DM -11.3 to -7.6); the two seeded Sobol points — included deliberately as unoptimized controls — lose to it (0.23064 and 0.23853), so the Optuna selection is worth 1–2 $\times$ 10^{-2} over a random draw. One expert's harvest is still completing; the bank aggregates below are computed on the nineteen complete experts over their full common row set (263,782 bars, 2003 onward), and the twentieth's inclusion is expected to move them only marginally.

The bank beats its members. The bank is aggregated two ways on the same trailing-125-bar tail with a 25-bar embargo, refreshed every 250 bars (the identical schedule to the linear tuners): a *selection* arm, which fills the next block from the lowest-tail-loss expert, and a *hedged* arm, which applies exponential weights $\propto \exp(-\eta \sum \ell)$ with $\eta = \sqrt{8 \ln K/L}$ — the standard weighted-average forecaster rate, no free constant. All tree rows are restricted to 2003 onward, since the menu was tuned on the pre-2003 dev span. The selection arm scores 0.21710 (DM -4.5 against the benchmark); the hedged arm scores 0.21296 (DM -15.3) — **better than every individual expert, including the best of them** (0.21796), a paired improvement of -0.0046 over the strongest member. The ordering (selection $<$ best expert $<$ hedge, with hedge -0.00413 better than selection at DM -5.3) is the paper's recurring tuning lesson in its purest form: causal selection on a short tail is expensive, and weighting is cheaper than selecting. The selection trajectory splits nearly evenly across families (296 XGBoost to 262 LightGBM boundaries) with era-specific leans (LightGBM in 2009 and 2013, XGBoost in 2018), and no expert dominates: the most-selected fills 22% of boundaries.

Table 5: The tree expert bank under the evaluation protocol (Section 4.2). Individual experts are scored on the full panel; the bank aggregates are computed on their common row set (263,782 bars, 2003 onward). DM t is against the OLS–HAR benchmark (0.22521); negative favours the arm.

	QLIKE	DM t
Best individual expert	0.21796	−11.3
Median completed expert	0.21969	—
Weakest Optuna-selected expert	0.22108	—
Sobol controls (2, unoptimized)	0.23064, 0.23853	—
`tree_tuned` (causal best-expert selection)	0.21710	-4.5
`tree_hedge` (exponential weights)	0.21296	-15.3

Comparison discipline against the linear ladder. The tree rows above are 2003+ by construction (263,782 bars); the linear arms are scored on 248,686–273,554 bars under their own row rules. Pooled levels across these row sets are not comparable (the campaign's row-set discipline of Section 4.2) — the hedged bank's 0.21296 and the linear champion's 0.21336 are separated by 2×10^{-5} *on different row sets*, which is exactly why the trees-versus-products verdict is reserved for the scorer's paired increments on common rows, registered as the block-ladder harvest completes (Appendix B.1). What is comparable as stated: the hedged bank's -0.00790 improvement over the benchmark at DM -15.3, and the bank-internal ordering above.

Where the nonlinearity lives. Three decompositions of the tree fits were computed in an earlier phase of this project.

Interaction order. A functional-ANOVA decomposition of a fitted depth-eight boosted tree attributes approximately 55% of its explainable variance to additive terms, 9% to pairwise interactions, and 36% to a higher-order remainder concentrated in no identifiable interaction. Within the additive part, the HAR block accounts for 96%.

Saturation at pairwise order. On the deployed residual-stack panel of the earlier build (n = 194,934; levels not comparable to the battery), an expressivity ladder holds the base model fixed and varies only the correction stage. A purely additive correction scores 0.12757. Adding bilinear pairwise interactions scores 0.12108. Nonlinear pairwise shapes score 0.12080. The deployed bagged additive-plus-pairwise stage (a GA2M in the sense of Lou et al. 2013) scores 0.12033. An Optuna-searched unrestricted-depth XGBoost correction stage scores 0.12094, which is worse than fitting no correction at all (0.12081). The ladder improves monotonically up to pairwise order; the unrestricted stage does not improve on the pairwise stages.

Distillation into products. On an earlier build of the same panel family, a penalized linear base scores 0.12516 and a residualized boosted-tree correction improves it to 0.12414. Adding to the base a single hand-written interaction — the HAR lags multiplied by a late-day session dummy, six columns — scores 0.12457, which recovers 58% of the tree correction's improvement. The full set of HAR-by-six-regime interactions scores 0.12453, or 62%. The session dummy in the recovered term marks the late-day bars at which the change of sign in HAR's persistence coefficient is documented in Section 3.

The tree experiments therefore yield a specification statement: **the panel's recoverable nonlinearity is a sparse set of pairwise products of columns the design already contains**.

5.3 Block-Diagonal Tikhonov

The block-diagonal Tikhonov of Section 4.4 is one solver with nested block penalties at a 24,000-bar window. The design is two blocks, $(\alpha_1, \alpha_2) = (1, 100)$: HAR barely penalized, the wide exogenous block shrunk hard. It scores QLIKE 0.22350, Diebold–Mariano -3.1 against the OLS–HAR incumbent.

Table 6: Block-diagonal Tikhonov under the evaluation protocol (Section 4.2), 24,000-bar window. DM t is against the OLS–HAR benchmark (0.22521); negative DM favours the arm.

	QLIKE	DM t	n
OLS on HAR + calendar (benchmark)	0.22521	—	273,554
Two-block $(1, 100)$	0.22350	-3.1	272,971

What the block-diag ridge carries runs in a linear estimator — one solver, two penalties, refit every bar. The next subsection reads that one-bar forecast against the last thirty minutes of the expiration-day option chain.

5.4 A Last-Bar Same-Day Option Reading

The forecasts of Table 6 are one-bar forecasts of realized variance. This subsection reads them against the last thirty minutes of the expiration-day chain, under the rules of Section 4.5: the nearest out-of-the-money call and put taken at the 15:30 ET mid, cash-settled at the close, with the position fixed by the sign of $s_t = \widehat{RV}_t - \mathrm{IV}^2_{30,t}$. Joining the chain to the seven forecast files leaves 871 expiration days common to all seven (3 January 2020 to 30 April 2024). Table 7 crosses every rule with every forecast on those days.

Table 7: Last-bar same-day option trade by position rule, 15:30 ET to the close (Section 4.5). One panel per rule; rows are the seven forecasts. One return per expiration day, on the 871 days common to all seven forecasts (3 January 2020 to 30 April 2024), filled at the 15:30 mid. $t = \sqrt{n}\,\text{mean/std}$ is computed on the daily R' and is not annualized; $\text{Sharpe}_{\text{ann}} = (\text{mean/std}) \times \sqrt{252}$ is the only annualized column. *ex. kurt* is Fisher excess kurtosis — the fourth standardized moment minus 3, so a Gaussian scores 0. The short-volatility rule takes no forecast, so its seven model rows are identical and it is shown as a single row.

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{ann}}$
Short volatility (always short)									
all models (no forecast)	871	0.020	1.127	-10.32	1.00	-2.31	10.90	0.53	0.28
$\text{sign}(s)$ *(± 1 on the sign of s)*									
baseline (HAR + calendar OLS)	871	0.071	1.125	-5.42	10.32	0.54	10.68	1.85	1.00
block-diag ridge	871	0.115	1.121	-5.42	10.32	0.96	10.56	3.03	1.63
LightGBM	871	0.094	1.123	-5.42	10.32	0.90	10.58	2.48	1.33
XGBoost	871	0.098	1.123	-5.42	10.32	0.84	10.60	2.58	1.39
lasso (causally tuned)	871	0.102	1.122	-6.84	10.32	0.44	10.75	2.68	1.44
lasso ($\alpha = 10^{-4}$)	871	0.130	1.120	-5.42	10.32	1.29	10.42	3.42	1.84
elastic net (causally tuned)	871	0.080	1.124	-6.84	10.32	0.44	10.71	2.11	1.14

Read the control first. Selling the package every day earns a mean 0.020 per expiration day at a standard deviation of 1.127 ($t = 0.53$, annualized Sharpe 0.28): a positive premium, at a size that 871 days cannot separate from zero. Putting the forecast in charge of *direction* is what pays. The rule $q = \text{sign}(s)$ earns between 0.071 and 0.130 per day across the seven forecasts (t between 1.85 and 3.42, significant at conventional levels for six of the seven), with the block-diagonal ridge at a mean of 0.115, $t = 3.03$, and an annualized Sharpe of 1.63. The forecast sits above the quoted implied variance — a long-volatility day — on 32 to 42 percent of days depending on the model.

The qualifications matter as much as the point estimates.

Where the level comes from. Two sample splits qualify the level rather than the sign. Excluding the first 64 trading days — the estimation warm-up, which happens to cover the February–March 2020 spike — *raises* the block-diagonal ridge portfolio to Sharpe 1.94 (Newey–West $t = 3.71$ on the remaining 807 days): the full-sample figure is dragged, not flattered, by the COVID quarter. Splitting instead at 16 May 2022, when SPXW listings became daily — a market-structure date, not a threshold chosen on the portfolio — the same portfolio scores Sharpe 1.13 ($t = 1.38$) before and 2.05 ($t = 2.86$) after: the direction edge is positive on both tapes but earns its significance on the daily-0DTE era.

Settlement pins carry a disproportionate share. The straddle expires worthless — $R = -1$ exactly — on 198 of the 871 days, 22.7 percent, whenever the close settles inside the one-strike window. The always-short control collects +1 on those days against -0.27 on all others; the $\text{sign}(s)$ portfolio averages +0.23 on them against +0.08 elsewhere, so pin days supply roughly half of its total return from under a quarter of the days. The portfolio monetizes a direction call and the pin premium together, and its P&L remains tilted toward days the index settles inside the strikes.

Seven rows are one bet. The seven daily portfolios are correlated at high pairwise levels, an effective count near one independent portfolio. Agreement down a column of Table 7 is evidence that the rule is insensitive to which forecast drives it — not seven independent confirmations of the rule.

The signal's content is a same-day sign, and the right test is the same-day one. The signal is fixed at 15:30 and the return settles at 16:00 on the same expiration day, so pairing s_t with R_t involves nothing observed after the decision. On that pairing the content is a shift in the conditional mean by sign: when the forecast exceeds the implied variance the package returns $+0.12$ on average, and -0.11 when it does not (difference 0.23, Newey–West $t = 2.9$ for the block-diagonal ridge; t between 1.9 and 3.3 across the seven forecasts), and the top third of days by the percentile rank of s_t beats the bottom third with t between 2.0 and 3.0 for every forecast. That is the sign(s) portfolio of Table 7, read as a regression, and Figure 1 shows it. A straight line is the wrong instrument for the same relation: the slope of R_t on the raw signal has no stable sign across forecasts (t from -0.5 to $+3.6$), because R has a point mass at -1 on 22.7 percent of days and a fat right tail, so the effect lives in the conditional means of the tails, which a sign or tercile split sees and a slope does not. The same rank is strongly related to the *magnitude* $|R_t|$ (t between 4.2 and 5.1), but that is largely the price level: once the day's implied variance enters the regression the rank's t falls to between 1.3 and 2.5 (implied variance itself $t \approx -3.8$) — a cheap straddle moves more per dollar of premium.

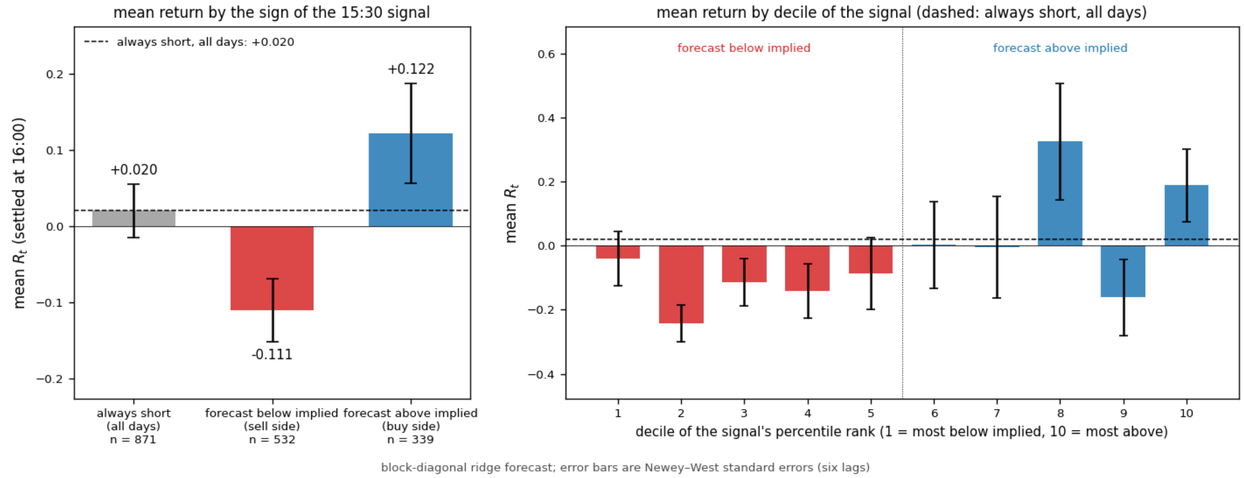

Figure 1: The 15:30 signal against the 16:00 settlement return, block-diagonal ridge forecast, 871 expiration days. Left: the mean settled return R_t on days the forecast sat below the quoted implied variance and on days it sat above, with Newey–West standard errors (six lags); the grey bar and the dashed line are the always-short portfolio over all days, the benchmark throughout. Right: the same mean by decile of the signal's percentile rank among all days up to and including t (decile 1 furthest below implied, decile 10 furthest above). The relation is a step at the sign with no trend within either half: the signal's content is which side of the quoted variance the forecast stands on, not how far.

Compounded at a causally estimated fraction, the $\mathrm{sign}(s)$ **portfolio grows; the control does not.** Betting the fraction of Equation (9) (mean $\hat{f}_t = 0.063$), the block-diagonal ridge $\mathrm{sign}(s)$ portfolio turns one unit of wealth into 36.8 over the sample (annualized log-growth $+1.04$), and into 12.2 at the half fraction. Its worst single day removes 49 percent of wealth at the full fraction and 25 percent at half — the case for the half path in one number. All seven $\mathrm{sign}(s)$ portfolios compound positively under the same procedure (terminal wealth 1.9 to 177), and the growth is uneven: 2021 and 2022 are negative years for the headline portfolio. The always-short control loses outright (terminal 0.67) — its mean is too small for the plug-in estimate, which is noise there. A defined-risk variant is parked and not reported here.

The trade as a whole is a demonstration under the stated constants of Section 4.5, not a trading claim: there is no delta hedge, no borrow, no market-impact model, and every fill is at the 15:30 mid, so a portfolio that receives the bid instead is a different portfolio. What survives the qualifications is worth stating plainly. At the last bar the quoted implied variance sits above the forecast on 61 percent of days; the sign of that gap carries a direction edge that is significant for six of the seven forecasts and strongest for the ridge this paper argues for; and the previous day's signal prices the size of the next settlement move rather than its direction.

6 Conclusion

On a quarter-million-bar intraday panel, we asked what exogenous information adds to a HAR backbone, what estimator can extract it, and what the nonlinearity is made of. The answers form a single chain. The exogenous panel carries real, statistically significant information — but the information is spread densely and weakly across hundreds of columns, so it is invisible to selection and to unpenalized joint fits, and recoverable only by shrinkage. The appropriate shrinkage is block-structured: almost none on the strong low-dimensional backbone, heavy and uniform on the wide exogenous block. The recoverable nonlinearity is small beside that information effect and is concentrated at pairwise order — products of columns the design already contains, gated on the session clock — a specification statement the tree fits deliver. The final model is ridge with a block-diagonal Tikhonov penalty, linear in its parameters throughout: two nested blocks, two fixed penalties, one exact rolling solve, and it outperforms the benchmark by a margin an order of magnitude beyond the fourth decimal place at strong paired significance. The paper closes by reading that one-bar forecast against the last thirty minutes of the expiration-day chain (Section 5.4): the sign of the gap between forecast and quoted implied is the tradable content — a ± 1 portfolio on $\mathrm{sign}(s)$ improves on the unconditional short — while the signal's magnitude tracks the size of the settlement move through the level of implied variance, not its direction.

Two findings sit beside the model rather than inside it, and may matter more broadly. First, the evaluation protocol is a modeling choice: the causal calibrated back-transform beats the retrospective Duan convention on every arm, and its adaptive second moment — not any conditional-mean device — is among the largest measured effects in the study. Second, causal hyperparameter tuning on short validation tails is repeatedly worse than well-chosen constants, because the price of a selection mistake is asymmetric across estimator families. Both findings are about the discipline surrounding the model, and both are measured, not assumed.

The limitations are stated where they bind. Results are computed on one asset complex and one loss; levels are comparable only within one preprocessing regime and one protocol, and arms that survive only under the legacy protocol are labeled as such in the appendix. The conditional-variance panel remains on the legacy scalar back-transform and has not been re-scored under the protocol used in the body. The natural extensions are equally concrete: promoting the mean-reverting second-moment refinement to the protocol (a re-score of every comparison); and model confidence sets over the feature sets once per-bar losses are persisted. None of these changes the chain above; each tightens a link in it.

A Master Table of Campaign Arms

Table 8 lists every arm of the unification campaign: one frozen panel (Section 3), one scoring protocol (Section B.1). QLIKE is the pooled loss under the evaluation protocol; Δ is the *paired* mean per-bar loss differential against the benchmark, and DM its Diebold–Mariano t-statistic, both computed on the row intersection of the two arms. **Protocol note.** Two row sets appear in this table: arms run under the original scoring protocol and arms run under the `oos_mult=2` protocol (chunks 9–99, 248,686 rows), which excludes the earliest out-of-sample period. Pooled QLIKE *levels* are not comparable across protocols — the excluded rows carry above-average loss, so `oos_mult=2` arms read mechanically lower. The Δ and DM columns are paired on common rows and are comparable throughout; level comparisons should be made only within a protocol. The gated two-block ridge and the *trail*-family arms are the `oos_mult=2` panel.

B The Back-Transformation

B.1 The Back-Transformation

Every model in this paper is fit on the preprocessed scale. Realized variance RV_t is diurnally adjusted and then passed through the square-root transform, so the regression target is $y_t = \sqrt{\widetilde{RV}_t}$, where $\widetilde{RV}_t$ denotes the diurnally adjusted variance at bar t. Evaluation is performed on the raw variance scale. This is because QLIKE — the primary loss — is a strictly consistent scoring rule for the conditional variance, not for any transform of it. A forecast issued in square-root space must therefore be mapped back to raw space before it is scored.

The naïve inverse squares the forecast and reverses the diurnal adjustment. Squaring does not commute with conditional expectation: by Jensen's inequality, $\mathbb{E}[\sqrt{X}]^2 \leq \mathbb{E}[X]$, with equality only in the degenerate case. Decompose the target as $y_t = \hat{y}_t + \varepsilon_t$, where $\hat{y}_t$ is the model's forecast of y_t and ε_t is a conditionally mean-zero innovation. Then

$$\mathbb{E}\left[y_t^2 \,\middle|\, \mathcal{F}\right] = \hat{y}_t^2 + \mathbb{E}\left[\varepsilon_t^2 \,\middle|\, \mathcal{F}\right], \tag{11}$$

where $\mathcal{F}$ is the conditioning information set. The conditional mean of the squared target exceeds the squared forecast by the conditional second moment of the square-root-scale innovation. A forecast that is well calibrated in $\sqrt{RV}$-space is, once squared, biased low in RV-space, and the size of the bias equals the residual variance of the model being evaluated.

Two facts about this bias determine the design. First, QLIKE penalizes under-prediction more heavily than over-prediction, and the bias of the uncorrected squared forecast is downward. Second, the bias equals each model's own residual variance, so it differs across models. The back-transform therefore includes an explicit second-moment term, and the estimator of that term is specified as part of the evaluation procedure.

The correction is a two-component causal calibration: a mean map re-estimated at every window boundary on the window before it, and a second moment seeded there and adapted bar by bar. Fix the chunked walk-forward grid: the backtest is split into contiguous evaluation windows indexed $k = 0, 1, \ldots$, tiling the panel in row order. The inputs for window k are three per-bar arrays over its bars: the forecasts $\{\hat{y}_t\}$ on the transformed (square-root, diurnally adjusted) scale, the diurnal baseline factors $\{B_t\}$, where B_t is the factor that reverses the Stage-1 adjustment at bar t, and the raw realizations $\{RV_t^{\text{raw}}\}$. The evaluation-scale target is the *unwinsorized* transformed realization, reconstructed exactly from the persisted raw target,

$$y_t^{\text{raw}} = \sqrt{RV_t^{\text{raw}}/B_t}. \tag{12}$$

Mean calibration. The forecast is passed through one affine map per window,

$$m_t = \hat{a}_k + \hat{b}_k\,\hat{y}_t, \qquad (\hat{a}_k, \hat{b}_k) = \arg\min_{a,\,b} \sum_{s\,\in\,k-1} \left(y_s^{\mathrm{raw}} - a - b\,\hat{y}_s\right)^2, \tag{13}$$

the Mincer–Zarnowitz regression of the evaluation-scale target on the forecast, estimated by ordinary least squares over the valid bars of window $k-1$ and applied to every bar of window k. The map absorbs the level and slope miscalibration the fit-scale construction leaves behind, including the attenuation induced by training on the winsorized target. For a window whose predecessor is missing from the harvest — a mid-panel gap, which a complete harvest does not contain — the map falls back to the identity $(\hat{a}_k, \hat{b}_k) = (0, 1)$; every such window is flagged in the harvest accounting.

Conditional second moment. Let $e_s = y_s^{\mathrm{raw}} - m_s$ denote the residuals against the calibrated mean as applied at their own bar. The second-moment term is adaptive. Window k opens at the seed

$$\hat{\sigma}^2_{(k)} = \frac{1}{N_{k-1}} \sum_{s\,\in\,k-1} e_s^2, \tag{14}$$

the mean of window $k-1$'s squared residuals, and within the window the term tracks the model's own recent error level causally,

$$\hat{\sigma}_t^2 = \lambda\,\hat{\sigma}_{t-1}^2 + (1-\lambda)\,e_{t-1}^2, \qquad \lambda = 0.6, \tag{15}$$

so the variance applied at bar t uses residuals through bar $t-1$ only. The decay sits on a broad, flat optimum: every value of λ between 0.4 and 0.6 scores within 10^{-4} of the best on the benchmark arm, and $\lambda = 0.6$ is the consensus choice across the block ladder's rungs. Positivity is trivial — a convex combination of squares — and the estimator does not extrapolate: no value of window k's forecasts can move it. The scalar of Equation (14) held fixed across the window — the protocol's first iteration — loses to the retrospective in-window Duan baseline on the benchmark (0.23353 against 0.23165) precisely because it cannot track the level; the adaptive form beats that baseline on every arm measured, while remaining strictly causal (benchmark 0.22519, paired ΔQLIKE -0.0065 at DM -5.7 against the Duan convention). For a window whose predecessor is missing from the harvest, the seed falls back to the in-window scalar mean of $(y_t^{\mathrm{raw}} - m_t)^2$; every such window is flagged in the harvest accounting.

Relation to the literature. Standard practice corrects the back-transform with a *static* scalar: the analytic log-normal factor $\exp(\hat{\sigma}_\varepsilon^2/2)$ of the log-RV literature, deployed since **?**'s lognormal–normal mixture and extended by Bekierman and Manner [2018] with the state variance of their state-space HAR, or Duan [1983]'s nonparametric mean of the exponentiated residuals. Whether the correction is worth applying at all is contested: Demetrescu et al. [2020] compare bias corrections for exponentially transformed forecasts, find none uniformly best, and recommend skipping the correction under high persistence; **?** reach a parallel conclusion in the multivariate case, finding parsimonious uncorrected forecasts best. The origin of the second moment's predictability is the realized estimator's own measurement error [Asai et al., 2012], whose variance scales with the volatility level.

Time-varying second moments are nonetheless standard *inside* transform-scale models, and each ingredient of the adaptive correction has a precedent. Volatility clustering in the residuals of realized-variance models motivates GARCH$(1,1)$ innovation variances in the HAR-GARCH of **?** — under a log-scale specification the retransformation factor becomes $\exp(m_t + h_t/2)$ in place of $\exp(m_t + \hat{\sigma}^2/2)$ — with **?** modeling log bipower variation and **?** the Spanish intraday electricity market with the same device. Inside log-variance GARCH models the Jensen gap is handled by construction, at a price: converting $\mathbb{E}[\log \sigma^2_{t+h} \mid \mathcal{F}_t]$ forecasts to $\mathbb{E}[\sigma^2_{t+h} \mid \mathcal{F}_t]$ requires the full h-step predictive distribution, approximated by Monte Carlo convolution of the one-step predictives [**?**]. The multivariate analogue of the second-moment correction is exact: **?** forecast the Cholesky factors of realized covariance matrices and derive the analytic second-moment correction from the forecast-error covariance. Outside finance, **?** shows that under heteroscedastic log-scale errors the scalar Duan [1983] factor is biased and must be replaced by a retransformation conditional on an estimated variance function; **?** find the conditional correction consistent but cumbersome enough to recommend sidestepping the back-transform entirely.

What we have not found is the correction used as a post-hoc *evaluation protocol*: a causal second-moment model of an arbitrary forecaster's own residuals, applied uniformly across heterogeneous model classes at evaluation time. In every precedent above the time-varying second moment is estimated jointly within one model's likelihood and serves that model's own forecasts. Equation (15) is the standard scalar correction with its level allowed to adapt causally — the static scalar is recovered at $\lambda = 1$. Equation (16) below is the covariance-stationary GARCH$(1,1)$ of **?**, with the intercept variance-targeted as in **?**, applied as an evaluation-layer second moment rather than estimated inside a forecast model's likelihood.

A mean-reverting refinement of the recursion. The volatility clustering that motivates **?** is present in this panel's residuals: the autocorrelation of e_t^2 is 0.15 at lag one and decays to zero within roughly one hundred bars. A covariance-stationary GARCH$(1,1)$ second moment [**?**],

$$\hat{\sigma}_t^2 = \hat{\omega}_k + \hat{A}_k\, e_{t-1}^2 + \hat{B}_k\, \hat{\sigma}_{t-1}^2, \qquad \hat{\omega}_k = (1 - \hat{A}_k - \hat{B}_k)\, \bar{e}_{k-1}^2, \tag{16}$$

targets the intercept to the previous year's mean squared residual [**?**], so the process reverts to a causally estimated level instead of integrating shocks permanently. The EWMA of Equation (15) is the integrated special case ($\hat{\omega} = 0$, $\hat{A} + \hat{B} = 1$) of **?**. The pair $(\hat{A}_k, \hat{B}_k)$ is re-selected at each year boundary by scored QLIKE on the previous year over a coarse grid ($\hat{A} \in \{0.02, 0.05, 0.1, 0.15, 0.2, 0.3\}$, $\hat{B} \in \{0.2, \ldots, 0.9\}$), and the recursion is seeded from the previous year's terminal state; every quantity applied in year k is a function of data through year $k-1$. Selections concentrate at persistent mean-reverting dynamics: $\hat{A} = 0.3$, the top of the grid, in most years, with $\hat{B}$ between 0.2 and 0.6, so $\hat{A} + \hat{B}$ lies between 0.5 and 0.9. Paired against the EWMA on identical bars, the refinement improves the benchmark's pooled QLIKE from 0.22072 to 0.21842 — ΔQLIKE -0.00230 at DM -9.10, the paired differential negative in 21 of 23 years — and improves the paper's best mean model from 0.21386 to 0.21044 — ΔQLIKE -0.00342 at DM -6.00, negative in 21 of 23 years.[1] That the gain is at least as large on the stronger mean model argues that the harvestable structure is volatility-of-volatility in the target, in the sense of **?**, rather than an artifact of a weak conditional mean. The deployed protocol nevertheless retains the EWMA of Equation (15):

[1] The shootout's EWMA comparator runs one continuous recursion seeded at the first scored year for a like-for-like race, so its pooled levels differ slightly from the frozen protocol's headline levels, which re-seed at every window boundary; the paired differentials are unaffected.

it has a single parameter on a flat optimum and no yearly selection step, and promoting the refinement would be a change of protocol in the sense of Section 4.2, requiring a re-score of every comparison that inherits it. Equation (16) is reported as an available upgrade, not adopted.

Calibration burn-in. The first evaluation window of each arm is the calibration burn-in, parallel to the feature construction's burn-in at the start of the panel: it estimates the two maps for its successor and is not scored — none of its bars enters any loss, any join, or any calibration diagnostic. Scoring begins at the second window. Every scored forecast is therefore computable at its own bar, with no exceptions.

The conditional-variance record. Two bar-conditional parameterizations of the second moment were implemented and rejected on measurement. An affine form, $\hat{\sigma}_t^2 = \max(\hat{c}_0 + \hat{c}_1 \hat{y}_t^2,\, 0)$, fit on the previous window's squared residuals, admits zero variance on quiet bars and lets the raw forecast collapse: under it, twenty bars carried 99% of one arm's pooled QLIKE. A log-linear form, $\hat{\sigma}_t^2 = \exp(\hat{d}_0 + \hat{d}_1 \hat{y}_t^2)\, \hat{S}$, with $\hat{S}$ the Duan [1983] retransformation factor for the log-back-transform, is structurally positive but extrapolates exponentially at bars whose $\hat{y}_t^2$ lies outside the previous window's fitted range: the second moment reached 10^{44}, inflating every arm's loss and collapsing the Diebold–Mariano differentials toward zero. A bar-conditional second moment therefore requires a model with controlled extrapolation, which is the reserved learned-correction slot for this section; the protocol uses the scalar of Equation (14).

A shrunk conditional second moment. The two rejected forms fail for the same reason: neither is anchored to the estimator that works. A third form removes that failure mode by construction. Let

$$z_t \;=\; \log\!\big((y_t^{\text{raw}} - m_t)^2\big),$$

and predict z_t by ridge regression on a standardized design, fit on the rolling window and refit at every bar. The penalty is selected causally on the same embargoed forward split used for every other hyperparameter in this paper, from a grid that includes $\alpha = \infty$; at that grid point the fitted value is the training-window mean of z and the assembled variance is *exactly* the window-level second moment of Equation (14) — the seed of the adaptive recursion (15). The window-level protocol is therefore nested as the infinite-shrinkage limit, and conditioning is adopted only where the forward split prefers it. At finite α the variance is

$$\hat{v}_t \;=\; \exp(\hat{z}_t)\, c_k, \qquad c_k \;=\; \frac{\text{mean of } (y^{\text{raw}} - m)^2 \text{ on window } k-1}{\text{mean of } \exp(\hat{z}) \text{ on window } k-1},$$

so each window's variance *level* is pinned to the scalar estimator's level and conditioning can only redistribute variance across bars within a window. No floor, clip, or additive constant appears anywhere in the construction.

Measured against the benchmark on the shared panel, the conditional protocol is the largest single effect in this study: applied to the benchmark's own residuals it improves QLIKE by +0.00365 at DM +3.5, and applied to the paper's best mean model by +0.00246 at +2.1. Both are an order of magnitude larger than any increment produced by changing the conditional mean. These conditional-variance runs were scored under the campaign's first protocol — the window-level scalar of Equation (14) held fixed within each window; under the adaptive recursion of Equation (15) part of this gain is absorbed by construction, since the conditional model and the recursion trade on the same level channel, and re-scoring the sidecar under the new protocol is reserved.

Which state carries the second moment. That result might be taken to show that the exogenous panel predicts forecast uncertainty; a control arm rejects that interpretation. Conditioning the variance on the backbone alone — the autoregressive ladder and calendar dummies, with no exogenous column — improves the benchmark by +0.00283 at DM +2.8, and beats the exogenous-state version head to head (-0.00082 at -2.9). The predictable part of the squared forecast error is therefore carried by the volatility level itself, which is what the realized-variance estimator's own measurement error scales with, and not by the exogenous state. The conditional second moment is a large and robust effect about heteroskedasticity in the target; it is not evidence that the exogenous buckets carry distributional information beyond the level, and this paper does not claim that they do.

Back-transform. Both members of the scored pair are mapped to the raw scale as

$$\widehat{RV}_t^{\text{raw}} = \left(m_t^2 + \hat{\sigma}_t^2\right) B_t, \qquad RV_t^{\text{raw}} = \left(y_t^{\text{raw}}\right)^2 B_t, \tag{17}$$

where the right-hand identity is Equation (12) inverted: the realization is the persisted raw target itself, never reconstructed from the winsorized fit target. The traditional scalar estimator of Duan [1983] is the same form with the identity mean map, a window-constant scalar, and in-window fit-scale residuals: the calibrated protocol differs from the classical Duan estimator in exactly three ways — the calibrated mean, the adaptive recursion of Equation (15) in place of the window-constant scalar, and causal residuals against that mean on the evaluation scale in place of the in-window fit-scale residuals (which see the scored window's own errors).

The full scoring procedure for one evaluation window is:

1. *Inputs.* The per-bar arrays $\{\hat{y}_t\}$, $\{B_t\}$, and $\{RV_t^{\text{raw}}\}$ over windows $k-1$ and k; reconstruct y^{raw} by Equation (12).

2. *Calibration.* Estimate $(\hat{a}_k, \hat{b}_k)$ by Equation (13) and the seed $\hat{\sigma}_{(k)}^2$ by Equation (14) on window $k-1$'s valid bars, with the stated fallbacks.

3. *Back-transform.* Map every bar of window k to the raw scale with Equation (17), producing the per-bar pairs $\left(\widehat{RV}_t^{\text{raw}}, RV_t^{\text{raw}}\right)$.

4. *Loss.* Evaluate QLIKE on the raw-scale pairs,

$$\text{QLIKE} = \frac{1}{N} \sum_{t=1}^{N} \left(\frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}} - \log \frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}} - 1 \right), \tag{18}$$

   where the mean is taken over the bars at which both members of the pair are positive; bars failing that condition are excluded from the mean.

Because $\hat{\sigma}_t^2$ is a convex combination of squares and $m_t^2 \geq 0$, the calibrated forecast $\widehat{RV}_t^{\text{raw}}$ is positive whenever B_t is and window $k-1$'s residuals are not identically zero; the positivity condition of the exclusion rule then binds only through the realization.

Three implementation choices define the estimator.

Window-level calibration, bar-level variance. The mean map $(\hat{a}_k, \hat{b}_k)$ is re-estimated at every window boundary from the window before it; the second moment opens at the previous window's scalar and then adapts bar by bar by Equation (15). On the raw scale the correction also varies through the baseline factor B_t. Relative to the squared calibrated forecast, the correction is largest at the bars where m_t^2 is smallest — the quiet bars — and largest in level immediately after the model's own recent errors have been large.

Causal, hence deployable. Every estimated quantity applied in window $k \geq 1$ is a function of data through window $k-1$ only, and windows are processed in strictly ascending order. The map is a quantity a deployed forecaster could have computed at t: the scored object is a real-time forecast, not a retrospectively corrected one. Scoring begins after the calibration burn-in, so there are no exceptions.

Residuals of the model itself, against the unwinsorized target. The residuals entering Equation (14) are the evaluated model's own out-of-sample forecast errors from the preceding window, measured against the unwinsorized evaluation-scale target of Equation (12) — not training-set residuals, not the residuals of any reference model, and not fit-scale residuals. The correction is therefore model-specific: an arm with larger residual variance receives a larger upward correction, so the correction is part of each arm's forecast, not a shared post-processing constant. With this choice the decomposition of Equation (11) holds for the evaluation target itself, tails included. One bias remains and is stated explicitly: the *coefficients* of each arm are estimated on the winsorized fit target, so the forecast $\hat{y}_t$ itself carries the fit-scale clipping. That is a modeling choice rather than a scoring defect; the mean calibration of Equation (13) absorbs its first-order attenuation, it is shared by every arm, and it cancels to first order in comparisons.

Protocol note. The scoring protocol's second-moment layer was upgraded on 2026-08-11 from the window-level scalar to the causal adaptive recursion of Section B.1 (Equation (15)). Rows harvested on Hoffman2 — the benchmark, the `oos_mult=2` panel, and every increment between them — are scored under the adaptive protocol. Rows whose chunks survive only on the decommissioned cluster (the original-protocol block ridges and the conditional second-moment panel) retain their *legacy* scalar-protocol levels; the two protocols shift QLIKE levels by roughly 5–8×10^{-3}, so legacy levels are not comparable to adaptive-protocol levels, while paired Δ and DM within each protocol remain valid. A mean-reverting GARCH$(1,1)$ refinement of the recursion (Equation (16)) is measured in Section B.1 but not adopted; promoting it would be a further change of protocol. The bucket and tuned grids completed their harvest under the current protocol on 2026-08-13 (the per-bucket attribution flipped accordingly: no bucket beats the benchmark unpenalized). The head-to-head table (Table 4) still carries a first-iteration scalar-protocol incumbent (0.23353) pending the lasso arm, launched 2026-08-13.

Table 8: All campaign arms under the frozen panel and scoring protocol.

Model	QLIKE	Δ	DM t
Benchmark			
OLS on HAR + calendar (benchmark)	0.22521	—	—

continued

Table 8 (continued)

Model	QLIKE	Δ	DM t
Single-bucket attribution (minimum-norm least squares)			
HAR + moments bucket	0.22761	+0.00240	+0.7
HAR + liquidity bucket	0.22934	+0.00413	+5.9
HAR + market bucket (equal-weighted)	0.22687	+0.00166	+4.6
HAR + market bucket (value-weighted)	0.22642	+0.00121	+3.2
HAR + sentiment bucket	0.22563	+0.00042	+1.4
HAR + implied-volatility bucket	0.22624	+0.00103	+3.9
HAR + volatility-demand bucket	0.22620	+0.00099	+3.4
HAR + all exogenous columns (joint)	0.22939	+0.00418	+6.3
Bucket designs under causally tuned ridge			
HAR + moments bucket	0.22407	-0.00114	-2.1
HAR + liquidity bucket	0.22817	+0.00296	+3.2
HAR + market bucket (equal-weighted)	0.22645	+0.00124	+2.7
HAR + market bucket (value-weighted)	0.22600	+0.00080	+1.7
HAR + sentiment bucket	0.22593	+0.00072	+1.7
HAR + implied-volatility bucket	0.22620	+0.00099	+2.9
HAR + volatility-demand bucket	0.22628	+0.00108	+2.4
HAR + all exogenous columns (joint)	0.22450	-0.00071	-1.1
Bucket designs under causally tuned elastic net (free mixing)			
HAR + moments bucket	0.22496	-0.00025	-0.4
HAR + liquidity bucket	0.22806	+0.00286	+3.2
HAR + market bucket (equal-weighted)	0.22647	+0.00126	+2.6
HAR + market bucket (value-weighted)	0.22634	+0.00114	+2.2
HAR + sentiment bucket	0.22624	+0.00103	+2.2
HAR + implied-volatility bucket	0.22646	+0.00125	+2.8
HAR + volatility-demand bucket	0.22632	+0.00111	+2.4
HAR + all exogenous columns (joint)	0.22570	+0.00049	+0.4
Penalized linear, full design			
Ridge, fixed $\alpha = 0.1$	0.22820	+0.00300	+4.5
Ridge, fixed $\alpha = 0.3$	0.22761	+0.00240	+3.7
Ridge, fixed $\alpha = 1$	0.22679	+0.00159	+2.6
Ridge, fixed $\alpha = 3$	0.22594	+0.00074	+1.2
Ridge, fixed $\alpha = 10$	0.22506	-0.00015	-0.3
Lasso, fixed $\alpha = 10^{-4}$	0.22382	-0.00139	-2.2
Ridge, causally tuned	0.22450	-0.00071	-1.1
Elastic net, causally tuned	0.22464	-0.00057	-0.6
Lasso, causally tuned	0.22566	+0.00045	+0.5
Block ridges			
Two-block ridge (stated penalties)	0.22350	-0.00171	-3.1
Two-block ridge (documented penalties)	0.22200	-0.00321	-4.7
Two-block ridge (per-block causal tuning)	0.22224	-0.00297	-5.6

continued

Table 8 (continued)

Model	QLIKE	Δ	DM t
Block ridges, `oos_mult=2` protocol (248,686 rows)			
Two-block ridge (gated rows, per-block causal tuning)	0.21431	-0.00330	-5.8
Conditional second-moment arms			
Benchmark, exogenous-state variance conditioning	0.22125	+0.00365	+3.5
Benchmark, backbone-only variance conditioning	0.22043	+0.00283	+2.8
Best model, exogenous-state variance conditioning	0.21613	-0.00229	-2.1

C Back-Transform Convention Sensitivity, All Arms

Table 9 reports every completed arm's pooled QLIKE under the three back-transform conventions of Section B.1, with within-convention ranks. The rank agreement statistics quoted there (τ against the second-order-adjustment ranking: none 0.682, traditional Duan 0.859) are computed over this table.

Table 9: Pooled QLIKE per arm under the three back-transform conventions (within-convention rank in parentheses).

Arm
OLS on HAR + calendar (benchmark)
HAR + moments bucket (OLS)
HAR + liquidity bucket (OLS)
HAR + market EW bucket (OLS)
HAR + market VW bucket (OLS)
HAR + sentiment bucket (OLS)
HAR + implied vol bucket (OLS)
HAR + vol demand bucket (OLS)
HAR + all features bucket (OLS)
Ridge, fixed $\alpha = 1$
Lasso, fixed $\alpha = 10^{-4}$
Ridge, causally tuned
Lasso, causally tuned
Elastic net, causally tuned
HAR + moments bucket (ridge, causally tuned)
HAR + liquidity bucket (ridge, causally tuned)
HAR + market EW bucket (ridge, causally tuned)
HAR + market VW bucket (ridge, causally tuned)
HAR + sentiment bucket (ridge, causally tuned)
HAR + implied vol bucket (ridge, causally tuned)
HAR + vol demand bucket (ridge, causally tuned)
HAR + all features bucket (ridge, causally tuned)
HAR + moments bucket (elastic net, causally tuned, free mixing)
HAR + liquidity bucket (elastic net, causally tuned, free mixing)
HAR + market EW bucket (elastic net, causally tuned, free mixing)
HAR + market VW bucket (elastic net, causally tuned, free mixing)
HAR + sentiment bucket (elastic net, causally tuned, free mixing)
HAR + implied vol bucket (elastic net, causally tuned, free mixing)
HAR + vol demand bucket (elastic net, causally tuned, free mixing)
HAR + all features bucket (elastic net, causally tuned, free mixing)
HAR + moments bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + liquidity bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + market EW bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + market VW bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + sentiment bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + implied vol bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + vol demand bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + all-features bucket (elastic net, causally tuned, free mixing, extended penalty grid)
Ridge, fixed $\alpha = 0.1$
Ridge, fixed $\alpha = 0.3$
Ridge, fixed $\alpha = 3$
Ridge, fixed $\alpha = 10$
Ridge, fixed $\alpha = 30$
Ridge, fixed $\alpha = 100$
Ridge, fixed $\alpha = 300$
Lasso, fixed $\alpha = 10^{-6}$
Lasso, fixed $\alpha = 10^{-5}$
Lasso, fixed $\alpha = 10^{-3}$
Lasso, fixed $\alpha = 10^{-2}$

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